

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Captive breeding, habitat creation & management - Set 1

Mark scheme

AQA spec 3.1.2.3.3-3.1.2.3.4; The living environment; textbook pp. 32-49

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • In-situ: conserving species in their natural habitat with the whole community; ex-situ: away from the natural habitat, eg zoos, botanic gardens, seed banks, captive breeding (1) • Needed when conservation in the wild is impossible or insufficient to prevent extinction (habitat gone, population too small, poaching) (1)	2
01.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Is the wild population threatened? (1) • Is there a genetically diverse captive population? (1) • Is in-situ conservation failing? (1) • Is keeping a captive population realistic (habitat size, food, relationships, cost)? (1) • Is release likely to be successful now or in future (suitable habitat, threats removed)? (1)	4
01.3	<i>2 marks for each point (partial credit for a partly correct point).</i> • Blue whale: habitat size far too large to provide; food (huge quantities of krill / plankton) cannot be supplied; cost (2) • Large blue butterfly: complex inter-species relationship - larvae are taken into ant nests where they overwinter feeding on ant grubs; requires specific ant species and thyme food plant (2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-34; AQA spec 3.1.2.3.3.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> • Very small gene pool so individuals are closely related and inbreeding is likely (1) • Harmful recessive genes are expressed when both parents carry them: some goslings had thin hair-like feathers with poor insulation so survival in cold weather was low (1) • A stud book records family trees so breeding is arranged between the least related individuals (often coordinated by a specialist zoo) (1) • Exchange of individuals / semen between zoos; avoid hybridisation with related species / subspecies (1)	4
02.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Breeding is triggered by stimuli (day length, light, temperature, food, body fat) which may be unknown / not provided (1) • Population interactions: mixing breeding and non-breeding individuals causes conflict; some need a choice of partners, others pair for life (1) • Breeding habitat needed: courtship display sites, nest sites, social groups (flamingos only breed in large groups - mirrors used) (1) • Stress / disturbance (1)	3
02.3	<i>1 mark for each point.</i> • AI: semen collected (and can be stored / cryopreserved for years) and inserted into the female, so animals need not be transported or meet; avoids dangerous mating (elephants, rhinos) and partner rejection (1) • Embryo transfer: a female of the endangered species (eg bongo) is treated with hormones to release many eggs, which are fertilised and implanted into surrogate females of a common related species (eland) (1) • So many offspring are produced per cycle instead of one calf per year, overcoming the shortage of breeding females (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 33-35; AQA spec 3.1.2.3.3.	
03.1	<i>1 mark for each point.</i> • Hard release: individuals released with no post-release support - suitable for species whose behaviour is instinctive and need no learned skills (insects, fish, amphibians, reptiles) (1) • Soft release: gradual release into larger areas with food provided while they learn to find food, avoid predators - needed for mammals and birds (1) • Problems faced: finding / recognising food and poisonous foods, hunting skills (captive-born predators never learned), avoiding predators, acceptance into wild social groups (1)	3
03.2	<i>Any 5 from the points below - 1 mark each, maximum 5.</i> • Large enough suitable habitat for a viable population (1) • Reliable food supply (1) • Low predation risk / original threat removed (1) • Suitable breeding sites (1) • Water (1) • Support of the local human population (1) • Official / legal protection of habitat and species (1)	5

Question	Marking guidance	Marks
03.3	<i>1 mark for each point.</i> - Tracking (radio / GPS tags, rings) shows survival, dispersal and breeding of released animals so the success of the project can be assessed (1) - Identifies causes of failure so future releases can be improved / extra support given (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 36-37; AQA spec 3.1.2.3.3.	
04.1	<i>1 mark for each point.</i> - The nucleus of a stem cell from the endangered species is transferred into an empty egg of a closely related species (southern white rhino) (1) - The embryo is implanted into a surrogate female of the related species which gives birth to a northern white rhino (1) - Very low success rates with wildlife so far; deep-frozen cells could allow clones even after the last individuals die (1)	3
04.2	<i>1 mark for each point.</i> - Micro-propagation: tissue culture producing many cell clusters from one plant, each grown into a genetically identical plant (1) - Black poplar: cuttings / pollarded stems of pure native trees are rooted to produce many identical trees (1) - Needed because the flowers are pollinated by introduced subspecies producing hybrids, so cloning maintains the pure gene pool (but no genetic diversity) (1)	3
04.3	<i>1 mark for each point.</i> - Stores seeds of wild plants (especially threatened species and crop wild relatives) so species are not lost if extinct in the wild (1) - Seeds collected worldwide; target of 10 000 seeds per species to maintain a diverse gene pool (1) - Stored dry and refrigerated / frozen in an underground reinforced vault to reduce risk of accident (1) - Seeds germinated periodically to check viability and replenish stocks; can be used for reintroduction / breeding (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 35-37; AQA spec 3.1.2.3.3.	
05.1	<i>1 mark for each point.</i> - Area must be large enough to support viable populations (breeding numbers high enough, gene pool large enough to avoid inbreeding); species higher up food chains need larger areas (lower density) (1) - Large areas contain a full range of resources (eg different fruiting trees / feeding areas through the year) (1) - Shape: the perimeter has edge conditions (a mix of neighbouring habitats); a compact shape maximises core habitat for species needing core conditions (1) - Some species benefit from edges (higher light) or small isolated habitats (ponds too small for predatory fish suit amphibians) (1)	4

Question	Marking guidance	Marks
05.2	<i>1 mark for each point.</i> - Varied topography: banks, lagoons, islands, ditches created with machinery give a range of water depths (emergent, floating-leaved, submerged plant zones) (1) - Islands provide roosting / nesting sites safe from terrestrial predators (foxes) (1) - Control of salinity (eg intertidal conditions for opossum shrimps feeding dunlin and avocet); scrapes for skylarks; different water depths for waders and ducks (1)	3
05.3	<i>1 mark for each point.</i> - Mobile species (birds, flying insects) colonise easily (1) - Less mobile species (many plants, amphibians, invertebrates) may not reach an isolated site so must be introduced; keystone species especially (1)	2
05.4	<i>1 mark.</i> Allows dispersal of surplus young, mixing of gene pools (less inbreeding) and recolonisation if a population dies out; avoids hazards (eg rope bridges for howler monkeys over roads) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 40-42; AQA spec 3.1.2.3.4.	
06.1	<i>2 marks for each point (partial credit for a partly correct point).</i> - Light: selective felling to create clearings for plants needing high light while retaining dense canopy for shade species; clearings also give warm sand / soil for lizard eggs; leave dead wood; woodland floor plants (bluebells) grow before the canopy leafs (2) - Dissolved oxygen: narrowing a slow river increases flow and turbulence so more oxygen dissolves, suiting trout and mayfly larvae; shallow water warms quickly for egg development (2) - Nutrients: removing fertile topsoil / not fertilising allows plants of nutrient-poor soils (eg sundew on acidic nitrate-deficient soils) that cannot compete on fertile ground; dung patches add local nutrients (2)	6
06.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> - Food: wildflower grasslands for seed-eating birds; providing food species (1) - Control of predation: island nest sites unreachable by foxes; trapping / culling / eradicating introduced rats (Scilly Isles - Manx shearwater and storm petrel breeding success increased) (1) - Control of competitors: removing rhododendron so wildflowers get light (1) - Pollination: a range of flowering plants with different flowering times so pollinators have food all season (1) - Seed dispersal species (hippos, elephants, monkeys) conserved; control of introduced pathogens (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 43-46; AQA spec 3.1.2.3.4, 3.1.3.1.	
07.1	<i>1 mark for each point.</i> - A community maintained at a stage before the climax by regular human activity (grazing, mowing, burning, coppicing) that deflects succession (1) - Chalk grassland was produced by centuries of sheep grazing; without it succession to scrub and woodland resumes (1)	2

Question	Marking guidance	Marks
07.2	<i>1 mark for each point.</i> • Grazing removes taller / dominant grasses and prevents scrub establishing (5% cover) so small plants are not shaded out (1) • Trampling creates bare ground for wildflower seed germination and warm patches for invertebrates / reptiles (1) • Dung creates small nutrient-rich patches supporting different plants and invertebrates; sheep eat short grasses and some bushes (1) • Mowing removes everything at once including plants grazers avoid, and removes flowers / seeds before setting; burning removes vegetation but also kills invertebrates and allows scrub regrowth; abandoned plot dominated by scrub (55%) so few species (1)	4
07.3	<i>1 mark for each point.</i> • Sheep eat short grass (some breeds eat scrub); ponies avoid flowers but eat bracken; cattle remove long rough grass (1) • Choice depends on the vegetation to be controlled and the plants to be protected (1)	2
07.4	<i>1 mark for each point.</i> • Removes mature heather bushes and young trees that grazers will not eat, creating open unshaded ground (1) • Heat stimulates dormant heather seeds to germinate so young heather re-establishes (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3.	
08.1	<i>1 mark for each point.</i> • Trees (hazel, oak) are cut to ground level every 3-30 years and allowed to regrow from the stump (1) • About 0.5 ha cut each year on a rotation (eg 8 years) so the wood is a patchwork of areas of different ages (1) • Each age has its own community: newly cut areas get high light for wildflowers and butterflies; older areas give dense cover for nesting birds / dormice (1) • Pollarding cuts above browsing height so deer / livestock cannot eat the regrowth (1)	4
08.2	<i>1 mark for each point.</i> • Wolves (top predator, exterminated by 1926) reduced deer populations and changed their behaviour (1) • Fewer young trees eaten so trees and riverside vegetation recovered (1) • More trees allowed beaver populations to increase; beaver dams increased wetland area (1) • Wetland species (birds, fish, invertebrates) became more common; overall biodiversity rose, though wolves cause problems for livestock farmers (1)	4
08.3	<i>1 mark for each point.</i> • Undesirable species (introduced predators / competitors, or natural succession species such as scrub, conifers in the New Forest heathland) reduce survival of the target species (1) • Removal restores suitable conditions (eg heathland for sand lizards and smooth snakes) or reduces predation / competition (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3-3.1.3.4.	